Evaluation runs: raft (raft-rs)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1,1	24	1323550	conforms	24.3	29	1 h, 250 GB	4	27	brain01, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, reorder 1 1 2 fifo, reorder 1 2 1 fifo, correct 1 2 1 fifo, correct 2 1 1 fifo, reorder 2 1 1 fifo, correct 2 2 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo.

stateful

stateful: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	24	1323550	conforms	23.9	29	1 h, 250 GB	brain01, 2026-11-02
1,1,2	24	490618796	violation*	9132.0	1916	48 h, 250 GB	brain01, 2026-11-05
1,2,1	24	12486268	conforms	258.6	62	48 h, 250 GB	brain01, 2026-11-05
2,1,1			🕓 (48 h)			48 h, 80 GB	pinky11, 2026-11-05

Not run (stateful). planned but never scheduled (grid points beyond the ladder): reorder 2 1 1 fifo, correct 2 2 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo, reorder 1 1 2 fifo, reorder 1 2 1 fifo.

offline

offline: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	24	1323550	conforms	22.2	32	1 h, 250 GB	0.1	22.2	0.0	merge	brain01, 2026-11-03
1,1,2	24	490618796	violation*	9029.0	1922	48 h, 250 GB	0.0	9029.0	0.0	merge	brain01, 2026-11-05
1,2,1	24	12486268	conforms	287.2	64	48 h, 250 GB	0.3	287.2	0.0	merge	brain01, 2026-11-05

Not run (offline). planned but never scheduled (grid points beyond the ladder): reorder 1 1 2 fifo, correct 1 3 1 fifo, reorder 2 1 1 fifo, correct 2 2 1 fifo.

refused (an input missing at run time): reorder 1 2 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo.

skipped: a smaller size had reached its limit: correct 2 1 1 fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	24	483541	conforms	1834.1	35876	10 min, 250 GB	0.1	1834.1	0.0	merge	brain01, 2026-11-03
1,2,1	24	6289273	conforms	40041.0	72120	48 h, 250 GB	0.3	40041.0	0.0	merge	brain01, 2026-11-06

Not run (cross offline, Java implementation against the Rust specification). left out by the table rules (ok): reorder 1 1 1 fifo.

skipped: a smaller size had reached its limit: correct 1 1 2 fifo, correct 2 1 1 fifo.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	24	1323550	conforms	24.0	187	10 min, 250 GB	4.1	22.2	1.8	merge	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, reorder 1 1 2 fifo, reorder 1 2 1 fifo, correct 1 2 1 fifo, reorder 2 1 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	24	483541	conforms	1952.2	37061	1 h, 250 GB		brain01, 2026-11-03

Not run (cross online, Java implementation (JMC) against the Rust specification export). left out by the table rules (ok): reorder 1 1 1 fifo.

left out by the table rules (timeout): correct 1 1 1 fifo, reorder 1 1 1 fifo.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	24	1323550	conforms	26.6	170	1 h, 250 GB		brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): reorder 1 1 1 fifo.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	483541	1323550	conforms	1834.2	35901	48 h, 250 GB	1834.1	22.2	0.1	merge	brain01, 2026-11-03
1,2,1	6289273	12486268	conforms	40041.0	72179	48 h, 250 GB	40041.0	287.2	0.0	merge	brain01, 2026-11-06

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): reorder 1 2 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo.

skipped: a smaller size had reached its limit: correct 1 1 2 fifo, correct 2 1 1 fifo.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1323550	483541	conforms	1834.1	35901	48 h, 250 GB	22.2	1834.1	0.0	merge	brain01, 2026-11-03
1,2,1	12486268	6289273	conforms	40041.0	72179	48 h, 250 GB	287.2	40041.0	0.0	merge	brain01, 2026-11-06

Not run (twins offline, Java implementation against its Rust twin). left out by the table rules (ok): reorder 1 1 1 fifo.

skipped: a smaller size had reached its limit: correct 1 1 2 fifo, correct 2 1 1 fifo.

twins online, Rust implementation (TraceForge) against the Java twin's export

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	483541	1323550	conforms	1878.6	35872	48 h, 250 GB		brain01, 2026-11-03

Not run (twins online, Rust implementation (TraceForge) against the Java twin's export). left out by the table rules (ok): reorder 1 1 1 fifo.

planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, correct 1 2 1 fifo, correct 2 1 1 fifo.